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9 September 1983
SUPERSEDING
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15 January 1973
(See 6.6)

MILITARY SPECIFICATION

UNIT HEATER, AIR-CIRCULATING, STEAM (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steam unit heaters for use in heating spaces on Naval ships.

1.2 Classification. Unit heaters shall be of the following classes and sizes, as specified (see 6.2.1).

Class 1 - Standard construction.
Class 2 - Nonmagnetic construction.
Size 11 - 18,000 British thermal units per hour (BTU/HR) capacity.
Size 12 - 30,000 BTU/HR capacity.
Size 13 - 46,000 BTU/HR capacity.
Size 14 - 74,500 BTU/HR capacity.
Size 15 - 116,000 BTU/HR capacity.
Size 16 - 186,500 BTU/HR capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter,

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SPECIFICATIONS

FEDERAL

FF-B-171	Bearings, Ball, Annular (General Purpose).
QQ-B-654	Brazing Alloys, Silver.
QQ-E-450	Electrodes, Welding, Covered: Mild Steel.
TT-P-645	Primer, Paint, Zinc-Chromate, Alkyd Type,
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917	Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-F-1183	Fittings, Tube, Cast Bronze, Silver-Brazing.
MIL-S-1222	Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 -	Plates, Identification.
MIL-E-15090 -	Enamel, Equipment, Light Gray (Formula No. 111).
DOD-P-15328 -	Primer (Wash), Pretreatment (Formula No. 117 for Metals).
MIL-M-17059 -	Motors, Alternating-Current, Fractional HP (Shipboard Use).
DOD-P-17545 -	Primer Coating, Alkyd-Red Lead Type, Formula No. 116 and Formula No. 116D.

STANDARDS

MILITARY

MIL-STD-167-1 -	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 -	Fabrication Welding and Inspection; and Casting Inspection; and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

2.1.2 Government publications. The following Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0900-LP-001-7000 -	Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

A 386 - Zinc Coating (Hot Dip) on Assembled Steel Products. (DoD adopted)
A 525 - Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process. (DoD adopted)
A 569 - Steel, Carbon (0.15 Maximum Percent), Hot Rolled Sheet and Strip, Commercial Quality. (DoD adopted)
B 6 - Zinc (Slab Zinc).
B75 - Seamless Copper Tube. (DoD adopted)
B 8 8 - Seamless Copper Water Tube. (DoD adopted)
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)
D 2092 - Preparation of Zinc-Coated Steel Surfaces for Painting.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Bulletin 20 - Test Code for Steam Unit Heaters.
Bulletin 300 - Test Code for Sound Rating.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 30 West University Drive, Arlington Heights, IL 60004.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence,

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Unless otherwise specified herein, the materials of construction shall be the same used in commercial unit heaters, except cast iron shall not be used in the construction of this unit heater.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment and material incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term ‘recovered materials” means materials which have been

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collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Class 1 unit heater. The class 1 unit heater shall be constructed of galvanized steel in accordance with ASTM A 525 with commercial thickness zinc coating or carbon steel in accordance with ASTM A 569, except that the heating element shall be made of copper or brass.

3.2.3 Class 2 unit heater. The class 2 unit heater shall be constructed of aluminum, brass or 300-series corrosion-resisting steel having a permeability of less than 2.0 after fabrication. The heating element shall be made of copper.

3.2.4 Corrosion protection. Brass, copper, 300-series corrosion-resisting steel, and galvanized steel are considered corrosion-resisting materials for this application. When the corrosion-resistance of the 300-series corrosion-resisting steel is degraded by fabrication processes, it shall be restored by heat treatment.

3.2.4.1 Parts fabricated from other than corrosion-resisting materials shall be protected against corrosion after fabrication by chemical electrolytic coatings such as plating, or the specified paints. The following methods are considered corrosion protection methods when properly applied:

(a) Hot-dipped galvanized shall be in accordance with ASTMA 386 with the spelter conforming to special highgrade in accordance with ASTM B 6.
(b) Electroplating with zinc shall be in accordance with ASTMB 633 followed by a phosphate treatment in accordance with method A of ASTM D 2092.
(c) Hot phosphoric or chromic acid treatment, or a coating of primer shall be in accordance with DOD-P-15328 followed by two coats of primer in accordance with TT-P-664.

3.2.4.2 Threaded fasteners and such fastening or fittings as may be used shall be of 300-series corrosion-resisting steel, or of steel coated in accordance with MIL-S-1222.

3.2.5 Dissimilar metals. Where dissimilar metal contacts can not be avoided, the materials shall be selected to minimize galvanic corrosion of the heater.

3.3 Desire and construction.

3.3.1 Maintenance. Maintenance of the equipment shall be demonstrated in accordance with 4.6.

3.3.2 General construction. The unit heater shall consist of a fan and motor, a twin header or double tube steam heating element and louvers, built as a single unit, provided with a fan guard, constructed for overhead mounting,

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ready for installation and connection to an alternating current power source and to the heating medium supply and return. The supply and return connections shall be on the same side of the unit.

3.3.2.1 Shock. The unit heater shall pass the high impact shock tests in accordance with MIL-S-901 for grade A, class I equipment.

3.3.2.2 Vibration. The unit heater shall be constructed such that no damage will occur or malfunction be caused either by internally excited vibrations, or by the environmental vibrations in accordance with MIL-STD-167-1 for frequencies up to, and including 33 hertz (Hz).

3.3.2.3 Noise. The unit heater by construction shall meet the sound power level in table I. The noise level shall be measured in accordance with 4.3.4.3.

TABLE 1. Sound power level, decibels re 10-12 watts.

Octave band (Hz)	63	125	250	500	1000	2000	4000	8000
Size 11	79	81	86	82	74	75	71	68
Size 12	82	84	87	84	77	77	72	68
Size 13	85	89	91	86	78	77	75	70
Size 14	87	90	92	89	82	79	76	71
Size 15	88	91	93	91	87	83	78	74
Size 16	92	102	95	93	88	84	81	76

3.3.3 Capacity rating. The capacity rating of the unit heater shall be at least equal to that as specified in table II, based on 25 pounds per square inch (lb/in^2) steam pressure, and 70 degrees Fahrenheit (°F) dry-bulb temperature entering the fan inlet. The temperature of the air leaving the heater shall not exceed that specified in table 11 by more than 3.0°F, and the air flow shall be within plus or minus 5 percent of the rated air flow as specified in table II.

TABLE II. Capacity.

Size	Capacity BTU/HR	Air flow Standard cubic feet/rein	Temperature, leaving, 'F
11	18,000	300	126
12	30,000	500	126
13	46,000	790	124
14	74,500	1330	122
15	116,000	2060	122
16	186,500	3310	122

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3.3.4 Physical dimensions. The overall dimensions of each unit heater shall not exceed those as specified on figure 1. The location and dimensions of the supply and return connections shall be as specified on figure 1.

3.3.5 Mounting dimensions. Mounting dimensions and mounting dimension tolerances of each unit heater shall be as specified on figure 1.

3.3.6 Heating element. The heating element shall be of the extended surface type consisting of at least 5/8 inch outside diameter (o.d.) seamless copper tubing expanded into fins, having twin headers or shall be of the double tube extended surface type consisting of at least 5/8 inch o.d. seamless copper tubing expanded into fins, an inner tube of 3/8 inch o.d. seamless copper tubing and a single header. The seamless copper tubes shall be in accordance with ASTM B 75, and shall have a minimum wall thickness of 0.032 for 5/8 inch o.d. tubes and 0.031 for 3/8 inch o.d. tubes. When used in the construction of the heating element, hairpin tubes shall have a minimum wall thickness of 0.032 inch throughout the tube. Brazed joints of the heating element shall be made with BCUP-5 silver brazing alloy in accordance with QQ-B-654.

3.3,6.1 Fins. Fins shall be of the smooth continuous plate type, not less than 0.012 inch thick and shall be uniformly spaced. Fins shall have rolled collars with tubes expanded into the collars to form a permanent bond.

3.3.6.1.1 The fin surface of the heating element shall be made of copper known commercially as “electrolytic tough pitch copper”.

3.3.6.2 Ferrules. Brass, copper or 300 series corrosion-resisting steel ferrules shall be provided where tubes pass through tube sheets. Tubes shall be expanded into ferrules to allow wear, to occur between the ferrules and tube sheets due to shipboard vibration and expansion and contraction of the tubes.

3.3.6.3 Header. The header(s) shall be made of copper and shall have a wall thickness equivalent to type L tubing in accordance with ASTM B 88 of equivalent size. The header(s) shall be, enclosed within the casing but not in the air stream. The supply and return connections to the header(s) shall be secured to the enclosure to relieve the header joints from stresses due to external pipe whip. The enclosure shall be secured to the casing by continuous welding or rivets spaced on not more than 2 inch centers.

3.3.6.3.1 The supply and return connections shall be fitted with type A unions in accordance with MIL-F-1183 with pre-inserted rings. Union sizes shall be as specified on figure 1.

3.3.6.4 When tested in accordance with 4.5.2.1, the heating element shall show no leakage.

3.3.7 Casing. The unit heater casing (when not aluminum or corrosion-resisting steel) shall be at least 0.059 inch thick when the coil face area is 1 square foot or less, and at least 0.104 inch thick when the coil face area exceeds 1 square foot. Casings constructed of aluminum shall be at least 0.102 inch thick when the coil face area is 1 square foot or less, and at least 0.140 inch thick when the coil face area exceeds 1 square foot. Casings constructed

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of 300 series corrosion-resisting steel shall be at least *0.047* inch thick when the coil face area is 1 square foot or less, and at least 0.076 inch thick when the coil face area exceeds 1 square foot.

3.3.8 Louvers. Each unit heater shall be provided with louvers. The louvers shall be constructed to permit adjustable directional air flow in the vertical plane and to prevent overheating the fan motor when the louvers are in the closed position.

3.3.9 Fan. The fan for the unit heater shall be of the propeller type, with at least three blades, complete with motor, motor mount, fan mounting panel and fan guard. The fan shall be quiet in operation and shall be dynamically balanced. Rotating parts of the fan shall be designed to have a factor of safety of not less than 8, based on the ultimate tensile strength of the material involved. Aluminum alloy hubs shall have a liner of 300 series corrosion-resisting steel. The fan blades shall be of such design that normal cleaning or dusting will not result in their being bent out of alinement. The fan blades shall be welded to the hub, or the blades and the hub tasted as a unit. The impeller shall be secured to the motor shaft by a key and by a castellated nut with a cotter pin.

3.3.9.1 Fan mounting. The unit heater fan shall be mounted on a tubular steel bracket with a nonadjustable motor mounting base. The bracket shall be permanently attached to the fan mounting panel. The tubular steel bracket, the motor mounting base and the fan mounting panel shall be sufficiently rugged *to* prevent distortion under shipboard conditions of vibration and shock. The fan motor shall be secured to the mounting base by positive means to prevent it from sliding under shipboard operating conditions. The fan mounting panel shall be secured to the casing by corrosion-resisting fasteners.

3.3.9.2 Fan guard. A fan guard shall be provided, which shall be securely fastened to the fan mounting panel and still permit ready removal. The fan guard shall incorporate the following requirements:

(a) prevent distortion under normal operating conditions.
(b) Protect personnel from injury.
(c) Minimize restriction of air flow.
(d) Minimize vibration when the fan is in operation.

3.4 Electrical equipment. The design of the electrical equipment shall be in accordance with the basic requirements of MIL-E-917.

3.4.1 Electric service. Fan motors shall be supplied for alternating current (a.c.) service. Fan motors 1/4 horsepower (hp), or less, shall be 115 volts, single phase, 60 Hz. Motors that exceed 1/4 hp shall be 440 volts, three-phase, 60 Hz.

3.4.2 Motors. Motors shall be service A in accordance with MIL-M-17059. Motors shall be designed for 65 degrees Celsius (°C) ambient temperature, continuous duty, constant speed and shall have class A insulation and spray-tight enclosure. Motors shall be equipped with normal internal fit ball bearing type 120 or type III, class 1, 2, or 7 in accordance with FF-B-171. A grounding circuit shall be established between the motor frame and the unit heater casing, and a separate ground shall be provided via a grounding wire

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encased in a synthetic rubber jacket. One end of the grounding wire shall be permanently installed to the unit heater casing; the other end (motor frame) shall have the capability of being easily connected or disconnected.

3.4.2.1 Motors, a.c. Motors, a.c., shall be designed for the following phase characteristics:

Current	Phase characteristics
Single phase	Split phase
Three phase	Squirrel-cage induction

3.5. Welding and brazing. The surfaces of all parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welding shall be in accordance with MIL-STD-278. Brazing procedures and qualification of welder (brazer) shall be in accordance with NAVSEA 0900-LP-001-7000.

3.5.1 Where zinc-coated steel is used for fabricating parts, the metallic zinc shall be removed from all joints and surfaces on which welds are to be deposited and for a distance of 1 inch from the expected toes of the welds. In areas where the metallic zinc cannot be removed and it is necessary to weld over the zinc-coated surfaces, electrode type MIL-G-6011 in accordance with QQ-E-450 shall be used.

3.6 Painting. Exterior surfaces of the unit heater, except those surfaces constructed of brass, copper or corrosion-resisting steel, shall be thoroughly cleaned and coated as follows:

(a) One coat of pretreatment coating in accordance with DOD-P-15328.
(b) One coat of zinc chromate primer in accordance with TT-P-645, or one coat of red lead primer in accordance with DOD-P-17545,
(c) Two coats of equipment enamel in accordance with class 2 of MIL-E-15090.

3.6.1 Interior surfaces. Interior zinc coated surfaces including welds and areas damaged by shearing, punching, bending, welding or any other fabricated process, shall receive a thorough wash and a hot phosphate acid treatment, or a pretreating coating in accordance with DOD-P-15328, prior to applying at least two coats of primer in accordance with TT-P-664 or DOD-P-17545.

3.7 Identification plate. Each unit heater shall be provided with an identification plate. The identification plate shall be style II and shall conform to type A, B, D, F or H in accordance with MIL-P-15024 and MIL-P-15024/5. The physical dimension of the identification plate shall not exceed the dimensions of a size number 10 plate in accordance with MIL-P-15024.

3.7.1 The identification plate shall contain the following information:

(a) Nomenclature.
(b) Size and capacity rating (see 3.7.1.1).
(c) National stock number (NSN).
(d) Component identification number (CID).

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(e) Contract or order number,
(f) Manufacturer's name and address.

3.7.1.1 The letters "NM" to designate that the unit is constructed of nonmagnetic material shall be added after the size designation on the class 2 unit heater.

3.8 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.8.1 Drawings. In addition to the drawing content required by the data ordering document (see 6.2.2), the following unique technical features shall be included:

(a) Methods and size of fastening and clearance for installation and servicing.
(b) Supplementary data necessary to permit shipyard installation without contractor's assistance.
(c) Performance data, curves:

(1) Leaving dry bulb (DB) temperature for the following conditions:

a. Entering DB temperature, 40 to 90"F at 5-degree intervals.
b. Steam supply pressure, 5 to 80 lb/in^2 at 15-lb/in2 intervals.

(2) Capacity in BTU/HR for the following conditions:

a. Entering DB temperature, 40 to 90°F at 5-degree intervals.
b. Steam supply pressure, 5 to 80 lb/in^2 at 15-lb/in^2 intervals.

(3) Condensate in pounds per hour (lb/hr) for the following conditions:

a. Entering DB temperature, 40 to 90°F at 5-degree intervals.
b. Steam supply pressure, 5 to 80 lb/in^2 at 15-lb/in^2 intervals.

3.8.2 Safety system check list. The contractor shall prepare a safety system check list in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.9 Workmanship. Unit heaters shall be free from defects that affect their appearance and operation. Casing tube sheets shall not show cracks due to punching or forming, and fin collars shall not show cracks after tubes are expanded into collars. Fin edges shall be free of burrs and shall not be bent. Casing corners shall be square, fan guard screen shall not be bent, and mounting bolts for directional louvers and fan assembly shall be in place and secured.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 First article inspection. First article inspection shall consist of the examination specified in 4.5.1 and the tests specified in 4.3.1 through 4.3.4.4 and 4.5.2.1 and 4.5.2.2.

4.3.1 Where a unit heater of a size larger than that submitted for tests has successfully passed the H.I. shock tests in accordance with 4.3.3, the H.I. shock may be extended by NAVSEA.

4.3.2 Unless otherwise specified (see 6.2.1), first article inspection will not be required on sizes of unit heaters that are identical to those units which have successfully passed the first article tests within a period of 5 years of the date of the contract or order. First article tests, assembled in manual form and certified by the inspector shall substantiate the fact that the tests have been conducted and that the specification requirements have been met.

4.3.3 High impact shock test. The sample unit heater shall be shock tested on the light weight machine, and the tests shall be as prescribed for type A in accordance with MIL-S-901. The fan of unit heater shall be operated at rated voltage and current, and a hydrostatic pressure of at least 100 lb/in^2 shall be maintained on the heating element of the unit heater during the shock tests. The test shall be conducted prior to tests specified in 4.3.4 and 4.5.2, and correction of damage which may have occurred during shock test shall not be performed prior to these tests. Evidence of fragmentation or missile effect of parts, deformation that will cause active interference between parts, or leakage of the heating element shall be cause for rejection.

4.3.4 Performance tests. Performance tests shall be conducted in the order in accordance with 4.3.4.1 through 4.3.4.4.

4.3.4.1 Vibration tests. The unit heater shall be subjected to the type I environmental vibration test in accordance with MIL-STD-167-1.

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4.3.4.2 Capacity, air flow rate and outlet temperature tests. Each size of unit heater shall be tested in accordance with AMCA Bulletin 20 except as follows:

(a) The steam supply at the heater shall be at a pressure not less than 23 lb/in^2 nor more than 27 lb/in^2 with a superheat of not less than 2°F nor more than 5°F.
(b) The mean outlet air temperature shall not exceed the temperature specified in 3.3.3 when the air inlet temperature is 70°F and the air flow rate does not exceed that specified in table II by more than 5 percent.
(c) Capacity and outlet temperature shall be corrected to 25 lb/in^2 steam pressure and 70"F inlet air temperature.

4.3.4.3 Sound test. The noise level of each unit heater with louvers and fan guards installed shall be established in accordance with AMCA Bulletin 300. Exceeding the level in any octave band is cause for rejection.

4.3.4,4 Motor test. The insulation resistance test and the dielectric test shall be conducted in accordance with MIL-M-17059.

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. Unit heaters of the same class and size offered for delivery at one time fill be considered a lot.

4.4.2 Sampling for examination. Sample unit heaters shall be selected at random from each lot in accordance with table III for the examination in accordance with 4.5.1.

TABLE III. Sampling for examination.

Lot size number of units	Sample size number of units	Defective units	
		Acceptance number	Rejection number
l t o 6	All	---	---
7 to 15	7	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	1	2

4.5 Quality conformance inspection.

4.5.1 Examination. Each of the sample unit heaters selected in accordance with 4.4.2 shall be examined and measured to verify compliance with the requirements of this specification, not involving tests. Examination shall be conducted in accordance with table IV. Any heater in the sample containing one or more defects shall not be offered for delivery, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

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TABLE IV. Classification of defects.

Categories	Defects
Critical:	
	None defined
Major:	
101	Size not as specified.
102	Incomplete, component parts missing or improperly assembled.
103	Material defective or not as specified.
104	Associated material noncompatible.
105	Fins not smooth continuous, plate type.
106	Fin collars and tube sheets cracked.
107	Supply and return header connections not as specified, pre-insert rings missing.
108	Welding incomplete, not free of cracks nonfusion heavy porosity, heavy undercut, slag inclusions.
109	Bolts, nuts and screws not tight, missing (parts shall be properly fastened and secured).
110	Fan assembly damaged or not as specified.
111	Drawings not followed.
112	Mounting dimensions not as specified.
113	Overall dimensions exceed the limits allowable.
114	Painting (as applicable), nonconforming.
115	Marking, manufacturer's name, not permanents illegible or not as specified.

4.5.2 Tests.

4.5.2.1 Leakage test. The heating element of each unit heater shall be tested at a hydrostatic pressure of 300 or 200 lb/in^2 air pressure with the heating element under water. Any unit heater in which there is evidence of leakage when tested shall not be offered for delivery.

4.5.2.2 Operational test. The fan assembly of each unit heater shall be tested at rated voltage and current to determine whether the fan will operate. Any unit heater in which the fan, or fan assembly fails to operate shall not be offered for delivery.

4.6 Maintenance demonstration. The first production unit shall be examined after testing, and the capability to maintain, disassemble and repair the unit shall be demonstrated to a Government representative. The demonstration shall be conducted utilizing the recommended tools and with other than expert mechanics. The maintainability demonstration shall include but not be limited to the following:

(a) After fan guards are removed, fan blades may be cleaned without requiring the removal of the fan assembly from the unit.
(b) The blades of fan may be cleaned without being bent out of alinement.

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(c) Heating element is cleanable without requiring removal from unit.
(d) Replacement of fan motor.
(e) Replacement of motor bearings.

4.7 Reports. The contractor shall prepare reports in accordance with the data ordering documents included in the contract or order (see 6.2.2) for the following:

(a) First article test (see 4.3).
(b) Shock tests (see 4.3.3).
(c) Vibration tests (see 4.3.4.1).
(d) Maintainability demonstration (see 4.6).

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5).

5.1 Preservation, packaging, packing and marking. Unit heaters shall be preserved, packaged, packed and marked level A or commercial as specified *(see 6.2.1)*.

6. NOTES

6.1 Intended use. The unit heaters are used as a heating system aboard Naval ships.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of specification.
(b) Class and size required (see 1.2).
(c) If first article inspection is required (see 3.1).
(d) Level of preservation, packaging, packing and marking required (see 5.1).
(e) On board repair parts required (see 6.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List *(CDRL)*, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data

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...not in the active inventory but are

specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.8	Microfilming of engineering documents, 35 mm and PCAM, preparation, microfilm aperture card	DI-E-20477	Microfilming of drawings on aperture cards - type I,. class 1 delivered to Portsmouth, NH, and type II, class 2 delivered to NAVSEA
3.8.1	Drawings, engineering and associated lists	DI-E-7031	Level 3, Design activity designation - contractor Drawing No - contractor
3.8.2	Safety system check list	UDI-H-26378	----
4.7	First article inspection report	UDI-T-4902	----
4.7	Equipment shock test report	UDI-T-23753	----
4.7	Vibration testing report	UDI-T-23762	----
4.7	Reliability maintainability assessment and demonstration reports	DI-R-3544	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000. 19L., Vol. 11, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should 'be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this" specification. This does not apply *to* specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been

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previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test) must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

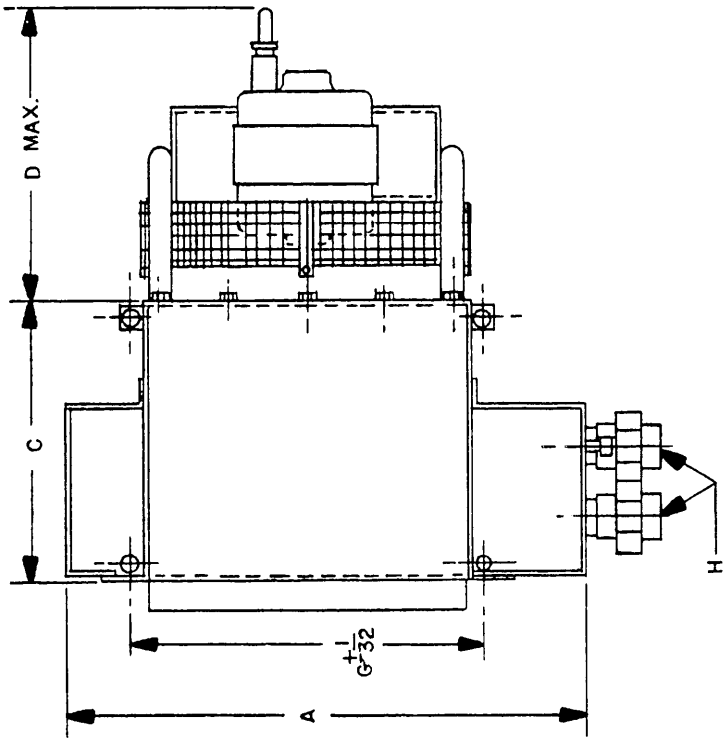
6.4 On board repair parts. When specified (see 6.2.1), on board repair parts for electric motors shall be furnished in accordance with the applicable motor specification specified herein.

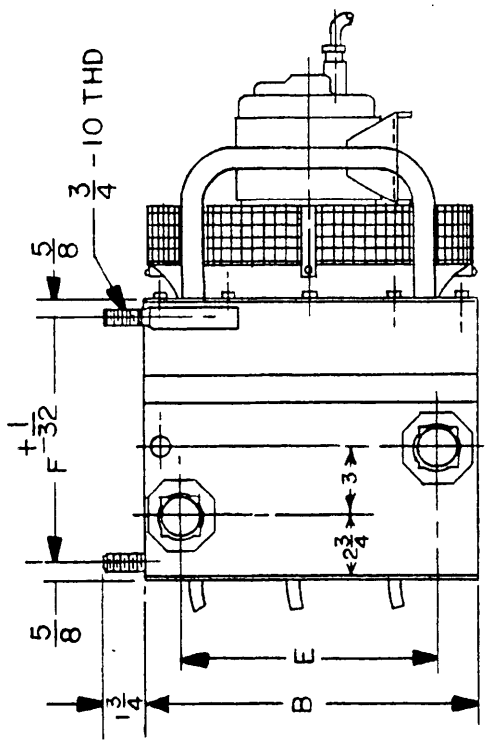
6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4520-N268)

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SH 7726 SIZE OF HEATER	DIMENSIONS IN INCHES, MAX.					FIXED DIMENSIONS		PIPE CONN. I.P.S.
	A	B	C	D	E	F	G	H
11	20	14	14	13	9	12-3/4	10	3/4
12	22	17	14	13	12	12-3/4	12	3/4
13	25	20	14	14	15	12-3/4	15	3/4
14	31	23	14	14	18	12-3/4	19	1-3/4
15	25	24	14	15	19	12-3/4	15	1-1/4
16	32	25	14	15	20	12-3/4	23	1-1/4

FIGURE 1. Physical dimensions of unit heater.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-U-17293D(SH)	UNIT HEATER, AIR-CIRCULATING, STEAM (SHIPBOARD USE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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